Practical No 04

a. Develop an Android application implementing **SQLite database CRUD operations**:.

Project Name: ServiceDemo

FileName: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:padding="20dp">
   <EditText
      android:id="@+id/etName"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:hint="Enter Name"/>
   <EditText
      android:id="@+id/etNewName"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:hint="Enter New Name"/>
   <Button
      android:id="@+id/btnInsert"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:text="Insert"/>
   <Button
      android:id="@+id/btnView"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:text="View Data"/>
   <Button
      android:id="@+id/btnUpdate"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:text="Update"/>
   <Button
      android:id="@+id/btnDelete"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:text="Delete"/>
   <TextView
      android:id="@+id/txtResult" android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:textSize="18sp"
      android:paddingTop="20dp"/>
</LinearLayout>
```

FileName: MainActivity.java

```
package com.example.sqllitecruddemo;


import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;


import androidx.appcompat.app.AppCompatActivity;


public class MainActivity extends AppCompatActivity {


    EditText etName, etNewName;
    TextView txtResult;


    Button btnInsert, btnView, btnUpdate, btnDelete;


    DBHelper dbHelper;


    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);


        etName = findViewById(R.id.etName);
        etNewName = findViewById(R.id.etNewName);
        txtResult = findViewById(R.id.txtResult);


        btnInsert = findViewById(R.id.btnInsert);
        btnView = findViewById(R.id.btnView);
        btnUpdate = findViewById(R.id.btnUpdate);
        btnDelete = findViewById(R.id.btnDelete);


        dbHelper = new DBHelper(this);


        // Insert
        btnInsert.setOnClickListener(v -> {


            String name = etName.getText().toString();


            dbHelper.insertData(name);


            Toast.makeText(
```

```
                this,
                "Data Inserted",
                Toast.LENGTH_SHORT
            ).show();
        });


        // View
        btnView.setOnClickListener(v -> {


            String data = dbHelper.viewData();


            txtResult.setText(data);
        });


        // Update
        btnUpdate.setOnClickListener(v -> {


            String oldName = etName.getText().toString();
            String newName = etNewName.getText().toString();


            dbHelper.updateData(oldName, newName);


            Toast.makeText(
                this,
                "Data Updated",
                Toast.LENGTH_SHORT
            ).show();
        });


        // Delete
        btnDelete.setOnClickListener(v -> {


            String name = etName.getText().toString();


            dbHelper.deleteData(name);


            Toast.makeText(
                this,
                "Data Deleted",
                Toast.LENGTH_SHORT
            ).show();
        });
    }
}
```

FileName: DBHelper.java

```
package com.example.sqllitecruddemo;


import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;


public class DBHelper extends SQLiteOpenHelper {


    public DBHelper(Context context) {
        super(context, "StudentDB", null, 1);
    }


    @Override
    public void onCreate(SQLiteDatabase db) {
        db.execSQL("CREATE TABLE student(name TEXT)");
    }


    @Override
    public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {
        db.execSQL("DROP TABLE IF EXISTS student");
        onCreate(db);
    }


    // Insert
    public void insertData(String name) {
        SQLiteDatabase db = this.getWritableDatabase();


        ContentValues values = new ContentValues();
        values.put("name", name);


        db.insert("student", null, values);
    }


    // View
    public String viewData() {
        SQLiteDatabase db = this.getReadableDatabase();


        Cursor cursor = db.rawQuery("SELECT * FROM student", null);


        StringBuilder data = new StringBuilder();


        while (cursor.moveToNext()) {
            data.append(cursor.getString(0)).append("\n");
```

```
        }

        cursor.close();

        return data.toString();
    }

    // Update
    public void updateData(String oldName, String newName) {
        SQLiteDatabase db = this.getWritableDatabase();

        ContentValues values = new ContentValues();
        values.put("name", newName);

        db.update(
                "student",
                values,
                "name=?",
                new String[]{oldName}
        );
    }

    // Delete
    public void deleteData(String name) {
        SQLiteDatabase db = this.getWritableDatabase();

        db.delete(
                "student",
                "name=?",
                new String[]{name}
        );
    }
}
```

b. Develop an Android application integrating SQLite database with ListView or RecyclerView to display, add, update, and delete records.

ProjectName: SQLiteCRUDDemo

FileName: Activity_main.xml

```
<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:padding="20dp">


   <EditText
      android:id="@+id/etName"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:hint="Enter Name" />


   <Button
      android:id="@+id/btnInsert"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:text="Insert" />




   <Button
      android:id="@+id/btnView"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:text="View Data" />
   <ListView
      android:id="@+id/listView"
      android:layout_width="match_parent"
      android:layout_height="wrap_content" />
</LinearLayout>
```

FileName: MainActivity.java

```
package com.example.sqlitecruddemo;


import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.Button;
import android.widget.EditText;
import android.widget.ListView;
import android.widget.Toast;


import androidx.appcompat.app.AppCompatActivity;


import java.util.ArrayList;


public class MainActivity extends AppCompatActivity {


    EditText etName;
    Button btnInsert, btnView;
    ListView listView;


    DBHelper dbHelper;


    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);


        setContentView(R.layout.activity_main);


        etName = findViewById(R.id.etName);
        btnInsert = findViewById(R.id.btnInsert);
        btnView = findViewById(R.id.btnView);
        listView = findViewById(R.id.listView);


        dbHelper = new DBHelper(this);


        // Insert Data
        btnInsert.setOnClickListener(v -> {


            String name = etName.getText().toString();


            dbHelper.insertData(name);


            Toast.makeText(
                    this,
```

```
                "Data Inserted",
                Toast.LENGTH_SHORT
            ).show();


            etName.setText("");
        });


        // View Data
        btnView.setOnClickListener(v -> {


            ArrayList<String> list = dbHelper.getData();


            ArrayAdapter<String> adapter = new ArrayAdapter<>(
                this,
                android.R.layout.simple_list_item_1,
                list
            );


            listView.setAdapter(adapter);
        });
    }
}
```

FileName: DBHelper.java

```
package com.example.sqlitecruddemo;


import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;


import java.util.ArrayList;


public class DBHelper extends SQLiteOpenHelper {


    public DBHelper(Context context) {
        super(context, "StudentDB", null, 1);
    }


    @Override
    public void onCreate(SQLiteDatabase db) {
        db.execSQL("CREATE TABLE student(name TEXT)");
    }


    @Override
    public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {
```

```
        // Database upgrade code goes here
    }


    // Insert Data
    public void insertData(String name) {
        SQLiteDatabase db = this.getWritableDatabase();


        ContentValues values = new ContentValues();
        values.put("name", name);


        db.insert("student", null, values);
    }


    // Get All Data
    public ArrayList<String> getData() {


        ArrayList<String> list = new ArrayList<>();


        SQLiteDatabase db = this.getReadableDatabase();


        Cursor cursor = db.rawQuery("SELECT name FROM student", null);


        while (cursor.moveToNext()) {
            list.add(cursor.getString(0));
        }


        cursor.close();


        return list;
    }
}
```

c. Develop an Android application to fetch JSON data from a web API and parse/display it in the Android application.

ProjectName: JSONDemo

FileName: Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>


<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:padding="20dp">


   <Button
      android:id="@+id/btnFetch"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:text="Fetch JSON Data" />


   <TextView
      android:id="@+id/txtResult"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:layout_marginTop="20dp"
      android:textSize="18sp"
      android:textColor="#000000" />


</LinearLayout>
```

FileName: MainActivity.java

```
package com.example.jsondemo;


import android.os.Bundle;
import android.widget.Button;
import android.widget.TextView;
import java.io.BufferedReader;
import java.io.InputStreamReader;
import java.net.HttpURLConnection;
import java.net.URL;
import org.json.JSONObject;
import androidx.appcompat.app.AppCompatActivity;




public class MainActivity extends AppCompatActivity {
```

```
Button btnFetch;
TextView txtResult;



@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);


    setContentView(R.layout.activity_main);



    btnFetch = findViewById(R.id.btnFetch);
    txtResult = findViewById(R.id.txtResult);



    btnFetch.setOnClickListener(v -> {
        fetchData();
    });
}



private void fetchData() {



    new Thread(() -> {



        try {



            // Web Service URL
            URL url = new URL("https://jsonplaceholder.typicode.com/posts/1");



            HttpURLConnection connection =
                    (HttpURLConnection) url.openConnection();


            connection.setRequestMethod("GET");
```

```
connection.setRequestMethod("GET");



BufferedReader reader = new BufferedReader(new InputStreamReader(
        connection.getInputStream()
)
);



StringBuilder result = new StringBuilder();



String line;



while ((line = reader.readLine()) != null) {
    result.append(line);
}



reader.close();



// JSON Parsing
JSONObject jsonObject =
        new JSONObject(result.toString());



String title =
        jsonObject.getString("title");



String body =
        jsonObject.getString("body");



// Display Data
runOnUiThread(() -> {
```

```
                txtResult.setText(
                        "Title: " + title +
                                "\n\nBody: " + body
                );
            });


            connection.disconnect();


        } catch (Exception e) {
            e.printStackTrace();
            runOnUiThread(() -> {
                txtResult.setText(
                        "Error: " + e.getMessage()
                );
            });
        }




    }).start();
  }
}
```

FileName: AndroidManifest.xml

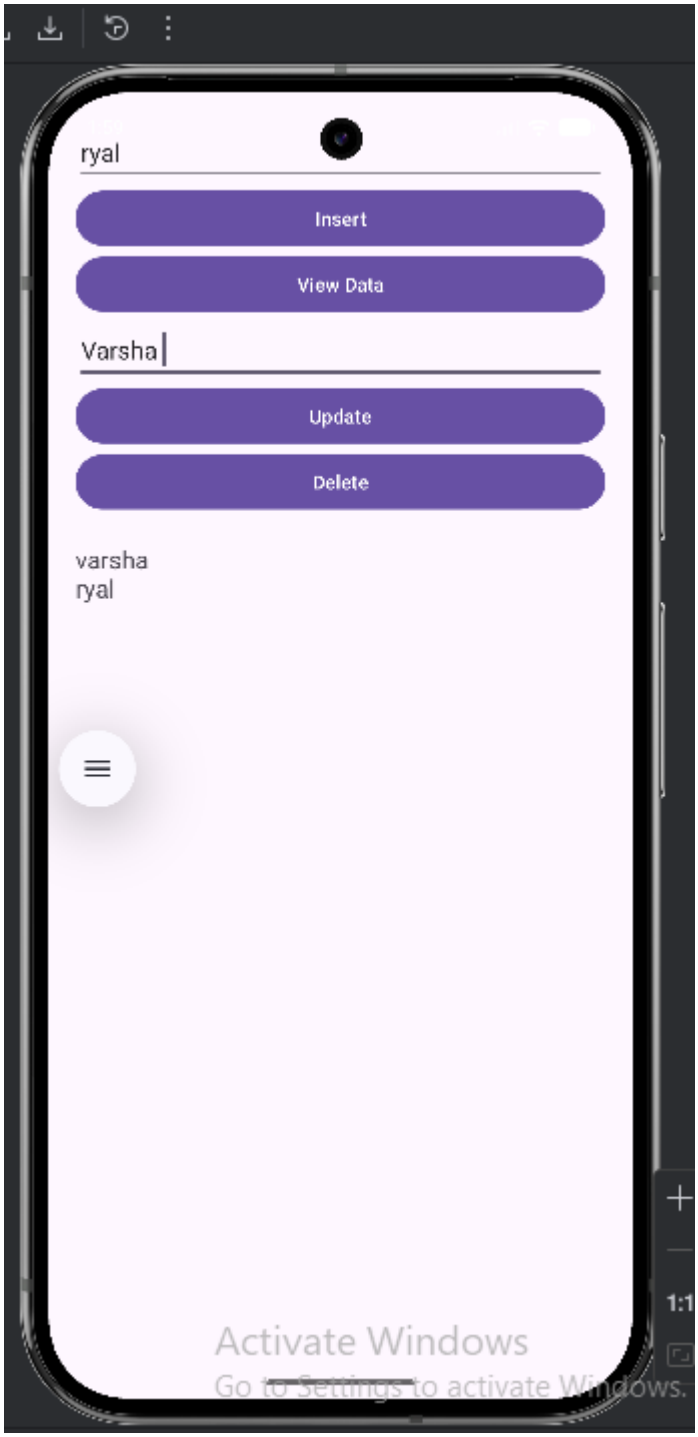
```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
  xmlns:tools="http://schemas.android.com/tools">
  <uses-permission android:name="android.permission.INTERNET"/>


  <application
    android:allowBackup="true"
    android:dataExtractionRules="@xml/data_extraction_rules"
    android:fullBackupContent="@xml/backup_rules"
    android:icon="@mipmap/ic_launcher"
    android:label="@string/app_name"
    android:roundIcon="@mipmap/ic_launcher_round"
    android:supportsRtl="true"
    android:theme="@style/Theme.JSONDEMO">
    <activity
      android:name=".MainActivity"
      android:exported="true">
      <intent-filter>
        <action android:name="android.intent.action.MAIN" />

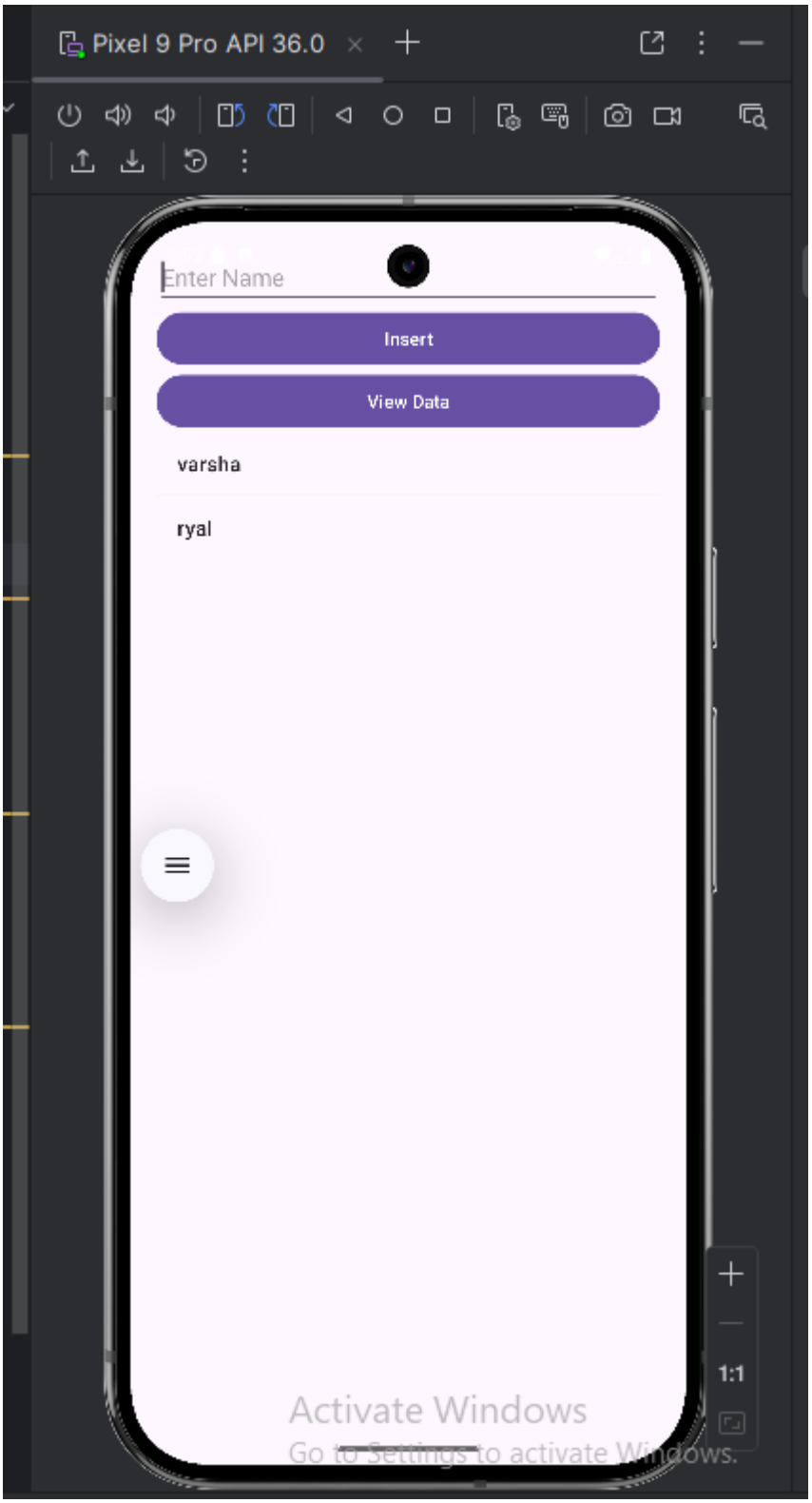
        <category android:name="android.intent.category.LAUNCHER" />
      </intent-filter>
    </activity>
  </application>

</manifest>
```

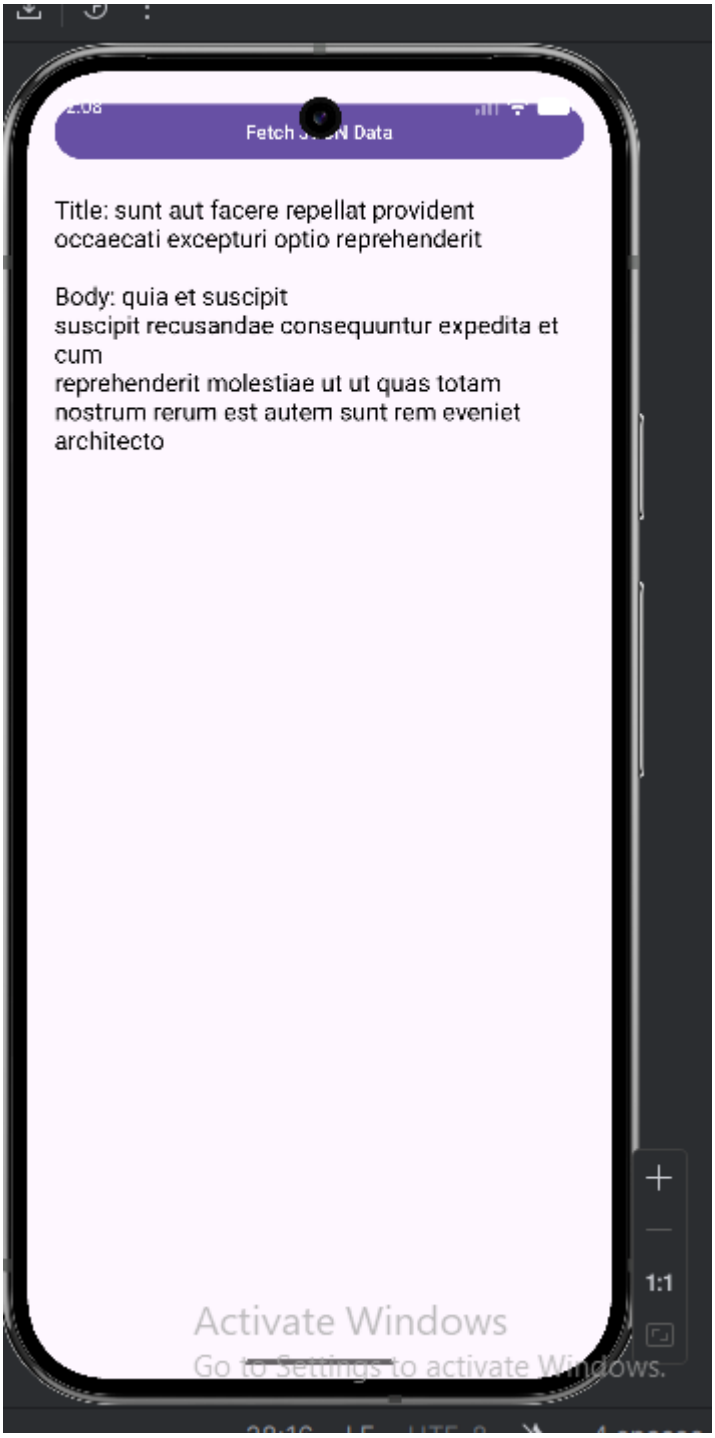
A) Output:


ryal
Insert
View Data
Varsha
Update
Delete
varsha
ryal
Activate Windows
Go to Settings to activate Windows.

B) Output:


Pixel 9 Pro API 36.0
Enter Name
Insert
View Data
varsha
ryal
1:1
Activate Windows
Go to Settings to activate Windows.

C) Output:


Fetch JSON Data
Title: sunt aut facere repellat provident occaecati excepturi optio reprehenderit
Body: quia et suscipit
suscipit recusandae consequuntur expedita et cum
reprehenderit molestiae ut ut quas totam
nostrum rerum est autem sunt rem eveniet architecto
1:1
Activate Windows
Go to Settings to activate Windows.